

Chapter 4

The Stochastic Integral

Let us fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, a filtration $(\mathcal{F}_t)$ that satisfies the usual conditions and a continuous local $(\mathcal{F}_t)$ -martingale M . Our goal is to define an integral process

$$t \mapsto \int_0^t f_t(\omega) dM_t(\omega)$$

for a large class of processes f using probabilistic arguments. If $M = B$ and $f_t(\omega) = f(t, B_t(\omega))$ for a sufficiently smooth function f , then this integral can be defined ω -by- ω as shown in the previous section. It was Itô's fundamental insight that one can generally *not* proceed in a pathwise way to define stochastic integrals. Instead, one needs to take a functional-analytic point of view, applying Hilbert spaces. As a warm-up we first introduce the stochastic integral w.r.t. Brownian motion. Subsequently, we show how to extend the main ideas to continuous local martingales.

4.1 Stochastic integration with respect to Brownian motion

We start with the stochastic integral w.r.t. Brownian motion. The definition of the stochastic integral is obvious for *elementary functions*.

Definition 4.1.1 *We call a process $\phi = (\phi_t)$ simple/elementary if it is of the form*

$$\phi_t(\omega) = \sum_j e_j(\omega) \mathbf{1}_{(t_j, t_{j+1}]}(t)$$

where $0 = t_0 < t_1 < t_2 < \dots$, where e_j is $\mathcal{F}_{t_j}$ -measurable and where $\sup_{i, \omega} |e_j(\omega)| < \infty$. The class of all such functions is denoted $\mathcal{L}_0$.

For any elementary function ϕ and every $T > 0$ we put^I

$$I_T(\phi)(\omega) = \int_0^T \phi_t(\omega) dB_t := \sum_{t_j \leq T} e_j(\omega) [B_{t_{j+1}} - B_{t_j}](\omega).$$

^IWe use again our convention on the sums along partitions and assume w.l.o.g. that the end point is an element of the partition. What we really mean is this: $I_T(\phi)(\omega) = \int_0^T \phi_t(\omega) dB_t := \sum_{t_j} e_j(\omega) [B_{T \wedge t_{j+1}} - B_{T \wedge t_j}](\omega)$.

The key insight is that the norm of an elementary function equals the norm of the stochastic integral. This is known as the *Itô isometry for elementary processes*.

Lemma 4.1.2 (*Itô isometry*) *Let ϕ be an elementary function. Then*

$$\mathbb{E} \left[\left(\int_0^T \phi_t(\omega) dB_t \right)^2 \right] = \mathbb{E} \left[\int_0^T \phi_t^2(\omega) dt \right].$$

PROOF: Let $\Delta_j B := B_{t_{j+1}} - B_{t_j}$ and fix $i \leq j \in \mathbb{N}$. Since e_i and e_j are measurable with respect to $\mathcal{F}_{t_i}$ and $\mathcal{F}_{t_j}$, respectively, we have for $i = j$ that

$$\begin{aligned} \mathbb{E}[e_i e_j \Delta_i B \Delta_j B] &= \mathbb{E}[\mathbb{E}[e_j^2 (\Delta_j B)^2 | \mathcal{F}_{t_j}]] \\ &= \mathbb{E}[e_j^2 \mathbb{E}[(\Delta_j B)^2 | \mathcal{F}_{t_j}]] \\ &= \mathbb{E}[e_j^2] (t_{j+1} - t_j) \end{aligned}$$

while for $i < j$ we obtain

$$\begin{aligned} \mathbb{E}[e_i e_j \Delta_i B \Delta_j B] &= \mathbb{E}[\mathbb{E}[e_i e_j \Delta_i B \Delta_j B | \mathcal{F}_{t_j}]] \\ &= \mathbb{E}[e_i \Delta_i B e_j \mathbb{E}[\Delta_j B | \mathcal{F}_{t_j}]] \\ &= 0. \end{aligned}$$

Hence

$$\begin{aligned} \mathbb{E} \left[\left(\int_0^T \phi_t(\omega) dB_t \right)^2 \right] &= \sum_{i,j} \mathbb{E}[e_i e_j \Delta_i B \Delta_j B] \\ &= \sum_j \mathbb{E}[e_j^2] (t_{j+1} - t_j) \\ &= \mathbb{E} \left[\left(\int_0^T \phi_t^2(\omega) dt \right) \right]. \end{aligned}$$

□

Construction of the Itô integral

The second fundamental insight is that the stochastic integral for simple processes is a martingale. More precisely, it belongs to the class $\mathcal{M}_2^{c,0}$ of all square integrable, continuous martingales that start in zero. The mapping $\langle \cdot, \cdot \rangle$ introduced in (3.3) defines a bilinear form on $\mathcal{M}_2^{c,0}$.

Proposition 4.1.3 *For any $\phi \in \mathcal{L}_0$ the integral process $I(\phi)$ belongs to $\mathcal{M}_2^{c,0}$.*

PROOF: By definition the integral process $I(\phi)$ starts in 0 and is adapted. Square integrability has already been established in the preceding lemma. The martingale property can be seen as follows: for all $s > t$,

$$\begin{aligned} \mathbb{E}[I_s(\phi) | \mathcal{F}_t] &= \mathbb{E} \left[\int_0^t \phi_u dB_u | \mathcal{F}_t \right] + \mathbb{E} \left[\int_t^s \phi_u dB_u | \mathcal{F}_t \right] \\ &= \int_0^t \phi_u dB_u + \mathbb{E} \left[\int_t^s \phi_u dB_u | \mathcal{F}_t \right] \quad (\text{by the definition of the integral}) \\ &= \int_0^t \phi_u dB_u + \mathbb{E} \left[\sum_{t \leq t_j < t_{j+1} \leq s} e_j \Delta_j B | \mathcal{F}_t \right]. \end{aligned}$$

Since e_j is $\mathcal{F}_{t_j}$ -adapted we see that $\mathbb{E}[e_j \Delta_j B | \mathcal{F}_t] = \mathbb{E}[e_j \mathbb{E}[\Delta_j B | \mathcal{F}_{t_j}] | \mathcal{F}_t] = 0$, and so

$$\mathbb{E}[I_s(\omega) | \mathcal{F}_t] = \int_0^t \phi_u dB_u = I_t(\omega).$$

□

So far, we have defined (pointwise for every $\omega \in \Omega$) a linear mapping

$$I : \mathcal{L}_0 \rightarrow \mathcal{M}_2^{c,0}.$$

In what follows we extend this mapping to a larger domain. We recall from Chapter 1 that the time integral of a progressively measurable process yields a progressively measurable process. In particular, the expected value of the time integral of the squared process is well defined.

Definition 4.1.4 Let $\mathcal{L}_2$ be the class of all jointly measurable and $(\mathcal{F}_t)$ -adapted functions $f_t(\omega) : [0, \infty) \times \Omega \rightarrow \mathbb{R}$ such that

$$\|f\|_{L_T^2(\lambda \otimes \mathbb{P})} := \mathbb{E} \left[\int_0^T f_t^2(\omega) dt \right] < \infty$$

for all $T > 0$. For any $f, g \in \mathcal{L}_2$, we set

$$d(f, g) := \sum_{n=1}^{\infty} 2^{-n} \min\{1, \|f - g\|_{L_n^2(\lambda \otimes \mathbb{P})}\}.$$

Proposition 4.1.5 The inclusion $\mathcal{L}_0 \subseteq \mathcal{L}_2$ is dense.

PROOF: Let $f \in \mathcal{L}_2$ and $T > 0$. We need to find a sequence $\{\phi^n\}$ of simple processes such that

$$\|f - \phi^n\|_{L_T^2(\lambda \otimes \mathbb{P})} < \varepsilon \quad \text{for all } n \geq n_0.$$

Then $d(f, \phi^n) < \varepsilon$ if T is large enough. To this end, we proceed in three steps.

- (i) Let us assume that f is uniformly bounded in (t, ω) and that the function $t \mapsto f_t(\omega)$ is a.s. continuous. Then the sequence of adapted, measurable processes

$$\phi_t^n(\omega) = \sum_j f_{t_j}(\omega) \mathbf{1}_{(t_j, t_{j+1}]}(t), \quad t \leq T$$

converges to $f \lambda \otimes \mathbb{P}$ -a.s. and in L_T^2 by dominated convergence.

- (ii) Let us now drop the assumption of continuity and assume only that f is uniformly bounded in (t, ω) and in addition that f is progressively measurable. Let

$$f_t^m(\omega) := m \int_{(t-1/m)^+}^t f_s(\omega) ds = m [K_t(\omega) - K_{(t-1/m)^+}(\omega)]$$

where

$$K_t(\omega) := \int_0^{t \wedge T} f_s(\omega) ds.$$

By Exercise 3 the uniformly bounded, continuous process K is progressively measurable, hence jointly measurable and adapted. By Step (i) there exist sequences $\{\phi^{m,n}\}$ of simple processes for every $m \in \mathbb{N}$ such that

$$\phi^{m,n} \rightarrow f^m \quad \text{in } L_T^2 \quad \text{as } n \rightarrow \infty.$$

Since f is bounded it is hence enough to prove that $f^m \rightarrow f$ $\lambda \otimes \mathbb{P}$ -a.s. Then L_T^2 -convergence follows from dominated convergence. To this end, let²

$$\mathcal{A} := \{(t, \omega) \in [0, T] \times \Omega : \lim f_t^m(\omega) = f_t(\omega)\}^c \in \mathcal{B}_T \otimes \mathcal{F}_T$$

and let

$$\mathcal{A}_\omega := \{t \in [0, T] : (t, \omega) \in \mathcal{A}\}.$$

By Fubini's theorem $\mathcal{A}_\omega \in \mathcal{B}_T$. Moreover, for all $t \in \mathcal{A}_\omega$

$$f_t^m(\omega) = m \int_{(t-1/m)^+}^t f_s(\omega) ds \rightarrow f_t(\omega).$$

By the fundamental theorem of calculus, since $f(\omega)$ is continuous this implies that $\lambda(\mathcal{A}_\omega^c) = 0$ and hence that

$$(\lambda \otimes \mathbb{P})(\mathcal{A}) = 0.$$

- (iii) Let us now assume that f is uniformly bounded but drop the assumption of progressive measurability. In this case, we do not know a priori that the continuous process

$$F_t := \int_0^t f_s ds$$

is adapted, hence progressively measurable by continuity. However, by Proposition 1.1.9 we do now that f has a progressively measurable modification g . Let

$$G_t := \int_0^t g_s ds.$$

Then G is progressively measurable. It is also a modification of F . To see this, we consider the measurable process $\eta_t(\omega) = \mathbf{1}_{\{f_t(\omega) \neq g_t(\omega)\}}$. By Fubini's theorem

$$\int_0^T \eta_t(\omega) dt = 0 \quad \text{a.s.}$$

Since

$$\{F_t \neq G_t\} \subseteq \left\{ \int_0^T \eta_t(\omega) dt > 0 \right\},$$

we conclude that $\{F_t \neq G_t\}$ is an event of measure zero. Since G is a modification of F and our filtration satisfies the usual conditions, we conclude that F is adapted. Since adaptedness and continuity imply progressive measurability, we can repeat verbatim the argument in (ii).

- (iv) Let f be progressively measurable. We can define bounded approximating functions by:

$$h_t^n(\omega) = \begin{cases} -n & \text{if } f_t(\omega) < -n \\ f_t(\omega) & \text{if } -n \leq f_t(\omega) \leq n \\ n & \text{if } f_t(\omega) > n \end{cases}.$$

□

²Notice that for a sequence of real-valued measurable mappings $\{g_n\}$ the set $\mathcal{A} := \{\lim g_n = 0\}$ is measurable because $\mathcal{A} = \bigcap_m \bigcup_k \bigcap_{n \geq k} \{|g_n| < 1/m\}$ and $\{|g_n| < 1/m\}$ is measurable.

Remark 4.1.6 Let us denote by $\mathcal{L}^*$ the subset of all (equivalence classes of) progressively measurable processes in $\mathcal{L}_2$. The proof of the preceding lemma actually shows that $\mathcal{L}_0 \subset \mathcal{L}^*$ dense.

The next result shows that the stochastic integral for simple processes extends uniquely to the set $\mathcal{L}_2$.

Proposition 4.1.7 The mapping $I : \mathcal{L}_0 \rightarrow \mathcal{M}_2^{c,0}$ extends uniquely to a continuous mapping

$$J : \mathcal{L}_2 \rightarrow \mathcal{M}_2^{c,0}$$

such that $I = J$ on $\mathcal{L}_0$ and such that J has almost surely continuous sample paths.

PROOF: Let us fix $f \in \mathcal{L}_2$ and a sequence of elementary processes $\{\phi^n\}$ that approximates f in the L^2 -sense.

- **EXISTENCE.** By the Itô isometry for simple processes, the sequence $\{I_t(\phi^n)\}$ forms a Cauchy sequence in $L^2(\mathbb{P})$ for any $t \geq 0$. By the completeness of $L^2(\mathbb{P})$, this shows that there exists a random variable $Y_t \in L^2(\mathbb{P})$ such that $I_t(\phi^n) \rightarrow Y_t$ in $L^2(\mathbb{P})$.
- **UNIQUENESS.** To show that the random variables Y_t are uniquely defined, let $\{\psi^n\}$ be another sequence that approximates f in the L^2 -sense. Then the isometry for elementary functions yields

$$\mathbb{E} \left[\left(\int_0^t \phi_s^n dB_s - \int_0^t \psi_s^n dB_s \right)^2 \right] = \mathbb{E} \left[\int_0^t (\phi_s^n - \psi_s^n)^2 ds \right] \rightarrow 0.$$

- **CONTINUITY OF THE SAMPLE PATHS.** We defined the process (Y_t) separately for each $t \geq 0$. We now prove that this process has a continuous modification. We know already that $I(\phi^n)$ is an $(\mathcal{F}_t)$ -martingale with continuous sample paths. Hence $I(\phi^n) - I(\phi^m)$ is also an $(\mathcal{F}_t)$ martingale with continuous sample paths so Doob's maximal inequality yields

$$\mathbb{P} \left[\sup_{0 \leq t \leq T} |I_t(\phi^n) - I_t(\phi^m)| > \varepsilon \right] \leq \frac{1}{\varepsilon^2} \mathbb{E}[|I_T(\phi^n) - I_T(\phi^m)|^2] \rightarrow 0 \quad \text{as } n, m \rightarrow \infty.$$

Hence we can choose a subsequence (n_k) such that

$$\mathbb{P} \left[\sup_{0 \leq t \leq T} |I_t(\phi^{n_k}) - I_t(\phi^{n_{k+1}})| > 2^{-k} \right] \leq 2^{-k}.$$

Thus, the Borel-Cantelli lemma shows that

$$\mathbb{P} \left[\sup_{0 \leq t \leq T} |I_t(\phi^{n_k}) - I_t(\phi^{n_{k+1}})| > 2^{-k} \text{ infinitely often} \right] = 0.$$

In particular, for almost all $\omega \in \Omega$, the sequence $\{I(\phi^{n_k})\}_{k \in \mathbb{N}}$ forms a Cauchy sequence in the space $C[0, T]$ of continuous real-valued functions on $[0, T]$, equipped with the supremum norm.³ Since this space is complete, there exists a limit process M that has almost surely continuous sample paths such that

$$I(\phi^{n_k}) \rightarrow M \quad \text{in } C[0, T] \text{ a.s.}$$

In particular, $I_t(\phi^{n_k}) \rightarrow M_t$ a.s. for all $t \in [0, T]$. This shows that

$$M_t = Y_t \quad \mathbb{P}\text{-a.s. for all } t \in [0, T].$$

³We recall that a sequence $\{x_n\}$ in a metric space is a Cauchy sequence if $\sum_n d(x_n, x_{n+1}) < \infty$.

The above arguments assumed a finite time interval $[0, T]$. In particular, the null-set $\mathcal{N}_T$ depends on T . To finish the proof, we just consider a sequence of compact time intervals $[0, N]$ for $N \in \mathbb{N}$ to obtain a stochastic process $(M_t)_{t \geq 0}$ with almost surely continuous sample paths on $(\bigcup_n \mathcal{N}_N)^c$. We define $J(f) = M$.

- **MARTINGALE PROPERTY.** We know that $\mathbb{E}[I_s(\phi^n)|\mathcal{F}_t] = I_t(\phi^n)$, $0 \leq t \leq s < \infty$. In particular,

$$\lim_{n \rightarrow \infty} \mathbb{E}[I_s(\phi^n)|\mathcal{F}_t] = J_t(f) \quad \text{in } L^2.$$

At the same time Jensen's inequality yields

$$\begin{aligned} \mathbb{E} \left[(\mathbb{E}[I_s(\phi^n)|\mathcal{F}_t] - \mathbb{E}[J_s(f)|\mathcal{F}_t])^2 \right] &= E \left[(\mathbb{E}[I_s(\phi^n) - J_s(f)|\mathcal{F}_t])^2 \right] \\ &\leq E \left[\mathbb{E}[(I_s(\phi^n) - J_s(f))^2 | \mathcal{F}_t] \right] \\ &= \mathbb{E}[(I_s(\phi^n) - J_s(f))^2] \\ &\rightarrow 0. \end{aligned}$$

- **CONTINUITY OF J .** This follows from the construction of the stochastic integral along with the Itô isometry for simple processes.

□

The process

$$J_t(f) =: \int_0^t f_s dB_s, \quad 0 \leq t < \infty$$

is called the *stochastic integral* (or Itô integral) of f with respect to Brownian motion. The following properties of the Itô integral are easily verified for elementary functions and hence for all $f \in \mathcal{L}_2$:

- (i) For any $0 \leq c \leq T$ we have $\int_0^T f dB = \int_0^c f dB + \int_c^T f dB$.
- (ii) Itô integrals are linear: For any constant c and all $f, g \in \mathcal{L}_2$ we have $\int (cf + g) dB = c \int f dB + \int g dB$.
- (iii) Expected values of Itô integrals are zero: $\mathbb{E}[\int f dB] = 0$.
- (iv) Itô integrals are adapted: The random variable $\int_0^t f dB$ is $\mathcal{F}_t$ -measurable.

The following exercise shows that the representation of Itô integrals is a.s. unique.

Exercise 35 Let $f, g \in \mathcal{L}_2$ and C, D be constants. Let

$$C + \int_0^T f dB = D + \int_0^T g dB.$$

Show that $C = D$ and that $f = g$ $dt \otimes \mathbb{P}$ -a.s.

Exercise 36 Prove that the following stochastic processes are martingales:

- (i) $X_t = e^{t/2} \cos(B_t)$,
- (ii) $X_t = (B_t + t) \exp(-B_t - \frac{t}{2})$

Exercise 37 (i) Prove directly from the construction of the Itô integral that

$$\int_0^t s dB_s = tB_t - \int_0^t B_s ds.$$

(ii) Show that the integral $\int B_s dB_s$ has finite first and second moments.

The Itô isometry

As an immediate corollary from the construction of the stochastic integral we obtain an isometry for functions belonging to the class $\mathcal{L}_2$.

Lemma 4.1.8 (Itô isometry) For every $f \in \mathcal{L}_2$ and any $t > 0$,

$$\mathbb{E} \left[\left(\int_0^t f_t dB_t \right)^2 \right] = \mathbb{E} \left[\int_0^t f_t^2 dt \right].$$

More generally, for any $f, g \in \mathcal{L}_2$ and any $0 \leq s \leq t < \infty$,

$$\mathbb{E} \left[\int_s^t f_u dB_u \int_s^t g_u dB_u \middle| \mathcal{F}_s \right] = \mathbb{E} \left[\int_s^t f_u g_u du \middle| \mathcal{F}_s \right].$$

PROOF: The first assertion follows from the construction of the stochastic integral and the Itô isometry for simple processes.

Using that $ab = \frac{1}{4}((a+b)^2 - (a-b)^2)$ it is enough to prove the second assertion for the case $f = g$. To this end, let $\{\phi^n\} \subset \mathcal{L}_0$ be such that $d(\phi^n, f) \rightarrow 0$. In particular,

$$\left\| \int_s^t \phi_u^n du - \int_s^t f_u du \right\|_{L^2(\mathbb{P})}^2 \rightarrow 0.$$

Let us then put

$$F_n := \int_s^t \phi_u^n dB_u \in L^2(\mathbb{P}) \quad \text{and} \quad F := \int_s^t f_u dB_u \in L^2(\mathbb{P}).$$

The first assertion yields

$$\|F_n - F\|_{L^2(\mathbb{P})}^2 = \mathbb{E} \left[\left| \int_s^t \phi_u^n dB_u - \int_s^t f_u dB_u \right|^2 \right] = \mathbb{E} \left[\int_0^t |\phi_u^n - f_u|^2 du \right] \rightarrow 0.$$

The lemma below implies that

$$\mathbb{E}[F_n^2 | \mathcal{F}_s] \rightarrow \mathbb{E}[F^2 | \mathcal{F}_s] \text{ in } L^1(\mathbb{P})$$

and

$$\mathbb{E} \left[\left(\int_s^t \phi_s^n ds \right)^2 \middle| \mathcal{F}_s \right] \rightarrow \mathbb{E} \left[\left(\int_s^t f_s ds \right)^2 \middle| \mathcal{F}_s \right] \text{ in } L^1(\mathbb{P}).$$

Now, the assertion follows from by the Itô isometry for simple processes from which we deduce that

$$\mathbb{E}[F_n^2 | \mathcal{F}_s] = \mathbb{E} \left[\left(\int_s^t \phi_s^n ds \right)^2 \middle| \mathcal{F}_s \right].$$

□

The preceding proof used the following well-known result from probability theory.

Lemma 4.1.9 Let f_n, f be square integrable random variables defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and let $\mathcal{G} \subset \mathcal{F}$ be a sub- σ -field. If $f_n \rightarrow f$ in L^2 , then

$$\mathbb{E}[(f_n)^2|\mathcal{G}] \rightarrow \mathbb{E}[f^2|\mathcal{G}] \quad \text{in } L^1.$$

PROOF: Using the Hölder inequality for conditional expectations

$$|\mathbb{E}[f_n^2 - f^2|\mathcal{G}]| \leq \mathbb{E}[|f_n + f||f_n - f||\mathcal{G}] \leq \sqrt{\mathbb{E}[|f_n + f|^2|\mathcal{G}]} \sqrt{\mathbb{E}[|f_n - f|^2|\mathcal{G}]}$$

An application of Jensen's inequality yields the desired L^1 -convergence:

$$\mathbb{E}|\mathbb{E}[f_n^2 - f^2|\mathcal{G}]| \leq \sqrt{\mathbb{E}[|f_n + f|^2]} \sqrt{\mathbb{E}[|f_n - f|^2]} \rightarrow 0 \text{ as } n \rightarrow \infty.$$

□

The following result is an immediate consequence of Lemma 4.1.8. It shows that if we subtract the covariation of an Itô integral from the square of the integral, then we get a martingale. This identifies the bracket process of Itô integrals!

Corollary 4.1.10 For any $f \in \mathcal{L}_2$ the following is a continuous martingale:

$$\left(\left(\int_0^t f_s dB_s \right)^2 - \int_0^t f_s^2 ds \right)_{t \geq 0}.$$

PROOF: Let

$$M_t := \int_0^t f_u dB_u.$$

Using Lemma 4.1.8, the martingale property of the stochastic integral and the orthogonality of martingale increments,

$$\begin{aligned} & \mathbb{E} \left[M_t^2 - \int_0^t f_u^2 du \middle| \mathcal{F}_s \right] \\ &= \mathbb{E} \left[(M_t - M_s)^2 - \int_s^t f_u^2 du + 2(M_t - M_s)M_s + M_s^2 - \int_0^s f_u^2 du \middle| \mathcal{F}_s \right] \\ &= M_s^2 - \int_0^s f_u^2 du \end{aligned}$$

□

Exercise 38 Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be C^1 everywhere and C^2 outside finitely many points $z_1, \dots, z_n$ and let $|g''(x)| \leq M$ for $x \notin \{z_1, \dots, z_n\}$. Let B_t be one dimensional Brownian motion. Prove that

$$g(B_t) = g(0) + \int_0^t g'(B_s) dB_s + \frac{1}{2} \int_0^t g''(B_s) ds.$$

Hint: Show that there exist functions $f_k \in C^2$ with bounded second derivative such that $f_k \rightarrow g$ uniformly, $f'_k \rightarrow g'$ uniformly, and $f''_k \rightarrow g''$ outside $\{z_1, \dots, z_n\}$. Then apply Itô to f_k and let $k \rightarrow \infty$.

Exercise 39 (Tanaka's formula and local time) What happens if we try to apply Itô's formula to $g(B_t)$ when $g(x) = |x|$? In this case g is not C^2 at $x = 0$. Instead, for any $\varepsilon > 0$ we modify $g(x)$ near $x = 0$ to $g_\varepsilon(x)$ as follows:

$$g_\varepsilon(x) = \begin{cases} |x| & \text{if } |x| \geq \varepsilon \\ \frac{1}{2} \left(\varepsilon + \frac{x^2}{\varepsilon} \right) & \text{if } |x| < \varepsilon \end{cases}.$$

(i) Apply Exercise [38](#) to show that

$$g_\varepsilon(B_t) = g_\varepsilon(B_0) + \int_0^t g'_\varepsilon(B_s) dB_s + \frac{1}{2\varepsilon} |\{s \in [0, t]; B_s \in (-\varepsilon, \varepsilon)\}|$$

where $|F|$ denotes the Lebesgue measure of the set F .

(ii) Prove that

$$\int_0^t g'_\varepsilon(B_s) 1_{(-\varepsilon, \varepsilon)}(B_s) dB_s = \int_0^t \frac{B_s}{\varepsilon} 1_{(-\varepsilon, \varepsilon)}(B_s) dB_s \rightarrow 0$$

in $L^2(P)$ as $\varepsilon \rightarrow 0$.

(iii) By letting $\varepsilon \rightarrow 0$ prove that

$$|B_t| = |B_0| + \int_0^t \text{sign}(B_s) dB_s + L_t(\omega), \quad (4.1)$$

where (the limit is taken in $L^2(\mathbb{P})$)

$$L_t = \lim_{\varepsilon \rightarrow 0} \frac{1}{2\varepsilon} |\{s \in [0, t]; B_s \in (-\varepsilon, \varepsilon)\}| = \lim_{\varepsilon \rightarrow 0} \frac{1}{2\varepsilon} |s \in [0, t] : |B_t| < \varepsilon|$$

and

$$\text{sign}(x) = \begin{cases} +1 & \text{if } x \geq 0 \\ -1 & \text{if } x < 0 \end{cases}.$$

The process L_t is called the local time for Brownian motion at 0 and [\(4.1\)](#) is the Tanaka formula for Brownian motion.

4.2 Stochastic integration w.r.t. continuous local martingales

We proceed to define the stochastic integral with respect to continuous martingales. Throughout, we fix a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ satisfying the usual conditions.

4.2.1 The space $\mathcal{M}$

In a first step we are going to define the stochastic integral for L^2 -bounded martingales. A martingale M is called L^2 -bounded if

$$\sup_{t \geq 0} \mathbb{E} M_t^2 < \infty.$$

In particular, an L^2 -bounded martingale is uniformly integrable.

Definition 4.2.1 The space of all L^2 -bounded, continuous martingales starting in zero is denoted $\mathcal{M}$.

For any $M \in \mathcal{M}$ the limit

$$\lim_{t \rightarrow \infty} M_t =: M_\infty$$

exists a.s. and in L^2 . Moreover, $(M_t)_{t \in [0, \infty]}$ is a martingale w.r.t. $(\mathcal{F}_t)_{t \in [0, \infty]}$ where $\mathcal{F}_\infty := \bigvee_{t \geq 0} \mathcal{F}_t$ (the fact that M extends to the time interval $[0, \infty]$ will be used repeatedly in what follows), and

$$M_t = \mathbb{E}[M_\infty | \mathcal{F}_t].$$

In particular, any element $M \in \mathcal{M}$ can be identified with the random variable M_∞ . Furthermore, we can define the bracket process $\langle M \rangle$ as a limit in probability. The following result shows that the bracket process introduces a Hilbert space structure on $\mathcal{M}$.

Theorem 4.2.2 1. *The space $\mathcal{M}$ is a real Hilbert space with scalar product*

$$\langle M, N \rangle_{\mathcal{M}} := \lim_{t \rightarrow \infty} \mathbb{E}[|M_t N_t|] = \mathbb{E}[|M_\infty N_\infty|].$$

and norm

$$\|M\|_{\mathcal{M}} := \mathbb{E}[M_\infty^2]^{1/2} \stackrel{3}{=} \mathbb{E}[\langle M \rangle_\infty]^{1/2}.$$

2. *The following norm on $\mathcal{M}$ is equivalent to $\|\cdot\|_{\mathcal{M}}$:*

$$\|M\| := \mathbb{E} \left[\sup_{t \geq 0} |M_t|^2 \right]^{1/2}.$$

3. *For $M \in \mathcal{M}$ the limit $\langle M \rangle_\infty := \lim_{t \rightarrow \infty} \langle M \rangle_t$ exists almost surely and*

$$\|M\|_{\mathcal{M}}^2 = \mathbb{E}[\langle M \rangle_\infty].$$

PROOF:

1. The second equality follows from L^2 -convergence $M_t^n \rightarrow M_\infty^n$ as $t \rightarrow \infty$. To show the Hilbert space property we need to prove that $\mathcal{M}$ is complete. To this end, let $\{M^n\}_{n \in \mathbb{N}}$ be a Cauchy sequence in $\mathcal{M}$. Then, $\{M_\infty^n\}_{n \in \mathbb{N}}$ is a Cauchy sequence in $L^2(\Omega, \mathcal{F}_\infty, \mathbb{P})$ by the definition of $\|\cdot\|_{\mathcal{M}}$ and hence the limit

$$M_\infty = \lim_{n \rightarrow \infty} M_\infty^n$$

exists in $L^2(\Omega, \mathcal{F}_\infty, \mathbb{P})$. As a result, it follows from Doob's L^2 -inequality that

$$\mathbb{E}[\sup_t |M_t^n - M_t^m|^2] \leq 4\mathbb{E}[|M_\infty^n - M_\infty^m|^2] \rightarrow 0 \quad \text{for } n, m \rightarrow \infty.$$

Hence there exists a subsequence (w.l.o.g. the original sequence) and a stochastic process $\bar{M}$ with continuous sample paths such that

$$M_t^n \rightarrow \bar{M}_t \quad \text{uniformly in } t \geq 0 \text{ a.s.}$$

Then $\bar{M}$ is adapted (using the usual conditions) and, using the above established L^2 -convergence,

$$\bar{M}_t = \lim_{n \rightarrow \infty} M_t^n = \lim_{n \rightarrow \infty} \mathbb{E}[M_\infty^n | \mathcal{F}_t] = \mathbb{E}[M_\infty | \mathcal{F}_t].$$

This shows that $\bar{M}$ is a continuous L^2 -bounded martingale.

2. Recalling that our martingale is defined for all $t \in [0, \infty]$, this follows from

$$\mathbb{E}[\sup_{t \in [0, \infty]} M_t^2] \leq 4\mathbb{E}[M_\infty^2] \leq 4\mathbb{E}[\sup_{t \in [0, \infty]} M_t^2].$$

This first inequality follows from Doob's maximal inequality; the second is trivial.

3. Exercise.

□

Exercise 40 Prove 3.) of the above theorem.

Remark 4.2.3 The proof of the first item of the above theorem provides the missing martingale argument in the proof Theorem [3.3.5](#) for bounded martingales.

4.2.2 Stochastic integration w.r.t. L^2 -bounded martingales

In this section we define the stochastic integral $h \cdot M := \int h dM$ with respect to a continuous martingale $M \in \mathcal{M}$. We proceed by analogy to Brownian motion. For a simple/elementary (left-continuous) process

$$h_t(\omega) = \sum_j h_j(\omega) \mathbf{1}_{(t_j, t_{j+1}]}(t)$$

where $0 = t_0 < t_1 < t_2 < \dots$, $h_j \in \mathcal{F}_{t_j}$ and $\sup_j |h_j| \leq C$ we define the stochastic integral $h \cdot M$ of h w.r.t. M as

$$h \cdot M := \left(\int_0^t h_s(\omega) dM_s \right)_{t \geq 0} := \left(\sum_{j=0}^{\infty} h_j [M_{t_{j+1} \wedge t} - M_{t_j \wedge t}] \right)_{t \geq 0}.$$

The class of all simple (left-continuous) processes is again denoted $\mathcal{L}_0$. The proof of the following lemma is similar to the Brownian case and hence left as an exercise. In particular, stochastic integrals of simple processes w.r.t. L^2 -bounded continuous martingales are martingales and their quadratic variation is given by the integral of the squared integrand with respect to the bracket of the martingale.

Lemma 4.2.4 For any elementary process h and any martingale $M \in \mathcal{M}$ the following holds:

1. $h \cdot M \in \mathcal{M}$
2. $\langle h \cdot M \rangle_t = \int_0^t h_s^2 d\langle M \rangle_s$.
- 3.

$$\|h \cdot M\|_{\mathcal{M}} = \mathbb{E} \left[(h \cdot M)_{\infty}^2 \right]^{1/2} = \mathbb{E} \left[\int_0^{\infty} |h_s|^2 d\langle M \rangle_s \right]^{1/2}.$$

Exercise 41 Prove the above lemma.

To extend the stochastic integrals to a larger class of integrands we put

$$\bar{\Omega} := \Omega \times [0, \infty), \quad \bar{\mathcal{F}} := \mathcal{F} \otimes \mathcal{B}.$$

For $M \in \mathcal{M}$ the associated *Doleans measure* $\mathbb{P}_M$ on $(\bar{\Omega}, \bar{\mathcal{F}})$ is defined by

$$\mathbb{P}_M(A) := \mathbb{E} \int_0^{\infty} \mathbf{1}_A(s, \omega) d\langle M \rangle_s.$$

The Doleans measure is a finite measure with total mass $\mathbb{E}[\langle M \rangle_{\infty}]$; see Theorem [4.2.2](#)(3.). We define

$$\mathcal{L}(M) := \mathcal{L}^2(\bar{\Omega}, \bar{\mathcal{F}}, \mathbb{P}_M)$$

as the Hilbert space of all measurable, adapted processes f such that

$$\|f\|_M := \|f\|_{L^2(\mathbb{P}_M)} < \infty.$$

Remark 4.2.5 We will be working with several subspaces of $\mathcal{L}(M)$ in what follows. We denote the corresponding spaces of equivalence classes using the letter “ L ”. That is, $L(M)$ is the space of equivalence classes of $\mathcal{L}(M)$, L_0 is the space of equivalence classes of $\mathcal{L}_0$ etc. In particular, $\|\cdot\|_M$ is a norm on $L(M)$ (but only a semi-norm on $\mathcal{L}(M)$).

In view of 3.) of the above lemma the stochastic integral defines an isometry

$$I : L_0 \rightarrow \mathcal{M}; \quad I(h) := h \cdot M$$

between the spaces L_0 and $\mathcal{M}$. An isometry from some metric space to a complete metric space can always be extended uniquely to an isometry on the completion of the first space. We denote the completion of L_0 in the Hilbert space $L(M)$ by

$$\overline{L}(M) := \overline{L_0}^{\|\cdot\|_M}.$$

We will then simply define the stochastic integral $h \cdot M$ for $h \in \overline{L}(M)$ as the image of the unique extension of our integral to a (linear) isometry from $\overline{L}(M)$ to $\mathcal{M}$ which we again denote by $h \cdot M$. In particular, the mapping $h \mapsto h \cdot M$ is linear.

It is of course of interest to have a more explicit description - or at least to identify a large subset - of the space $\overline{L}(M)$.

Remark 4.2.6 It is definitely not true (in general) that $\overline{L}(M) = L(M)$. If $L'(M)$ denotes the closed subspace of $L(M)$ consisting of all progressive processes in $L(M)$ (more precisely: consisting of all equivalence classes containing some progressive process) then $L'(M)$ contains L_0 (since left continuous adapted processes are progressive) and therefore also its completion $\overline{L}(M)$. We emphasize that an equivalence class of $L(M)$ may well contain both progressive and non-progressive processes: just think of the extreme case in which $M \equiv 0$. Then $\mathbb{P}_M = 0$ and all processes in $\mathcal{L}(M)$ are equivalent.

If $\langle M \rangle$ is absolutely continuous w.r.t. Lebesgue measure, then a straightforward extension of the arguments given in the proof of Proposition 4.1.5 yields the following result.

Proposition 4.2.7 If $\langle M \rangle$ is absolutely continuous w.r.t. Lebesgue measure, then L_0 is dense in the subset of all measurable, adapted processes in $L(M)$.

Predictable integrands

We are now going to show that any predictable process in $\mathcal{L}(M)$, that is, any process in $\mathcal{L}(M)$ that is measurable w.r.t. the predictable σ -algebra on $\mathcal{B} \otimes \mathcal{F}$, can be approximated by simple functions with respect to the semi-norm $\|\cdot\|_M$.⁴ In principle, we know already from Exercise 2(ii) that this is indeed true but we did not work with the Doleans measure there. The following result shows that the use of the Doleans measure does not alter anything. The proof uses the following vector version of the monotone class theorem.

Monotone Class Theorem (Vector version) Let $\mathcal{D}$ be a Dynkin system and $\mathcal{H}$ be a set of functions from Ω to $\mathbb{R}$ that satisfies the following properties:

⁴Recall that the predictable σ -algebra on $\mathcal{B} \otimes \mathcal{F}$ is contained in the progressive one and is generated by the processes in $\mathcal{L}_0$, that the predictable σ -algebra is the smallest σ -algebra w.r.t. which all left-continuous processes are measurable and that the predictable σ -algebra is contained in the progressive σ -algebra; cf. Exercise 2.

- $\mathcal{H}$ is a vector space
- if $D \in \mathcal{D}$, then $\mathbf{1}_D \in \mathcal{H}$
- $\mathcal{H}$ is closed with respect to bounded increasing limits

Then $\mathcal{H}$ contains all bounded $\sigma(\mathcal{D})$ -measurable functions.

Proposition 4.2.8 *Let $M \in \mathcal{M}$ and $h \in \mathcal{L}(M)$ be predictable. Then there exists a sequence of simple processes $\{h^n\}$ such that $\|h - h^n\|_M \rightarrow 0$ as $n \rightarrow \infty$. In particular,*

$$h \cdot M \quad \text{exists for any predictable } h \in \mathcal{L}(M).$$

PROOF: Let $\mathcal{R}$ be the class of all bounded, predictable processes that can be approximated as claimed. Then $\mathcal{R}$ is a vector space that contains all constant processes as well as the family of indicators

$$A \times (s, t], \quad A \in \mathcal{F}_s.$$

Since this family generates the predictable σ -field the above monotone class theorem states that $\mathcal{R}$ is equal to the set of all bounded predictable functions provided we can show that $\mathcal{R}$ is closed with respect to limits of increasing, nonnegative and uniformly bounded elements of $\mathcal{R}$. To show this, take a sequence of non-negative processes g^n from $\mathcal{R}$ such that $g^n \uparrow g$ where g is bounded. Being the pointwise limit of predictable processes g is also predictable. Further, we have $\|g^n - g\|_M \rightarrow 0$ by dominated convergence and it follows from the usual “triangle trick” that $g \in \mathcal{R}$. This shows that any bounded predictable process can be approximated as claimed. To complete the proof, let h be as in the proposition and define $h^{(N)} := h \mathbf{1}_{|h| \leq N}$. Then $h^{(N)} \in \mathcal{R}$ and $\|h^{(N)} - h\|_M \rightarrow 0$ by dominated convergence. Again the triangle trick shows that the proposition holds for h . $\square$

Progressive integrands

The following proposition shows that progressively measurable processes can also be approximated by simple processes, at least when restricted to finite time intervals.

Proposition 4.2.9 *Let A be an adapted, continuous, increasing process starting in zero (think of A as $\langle M \rangle$) such that $\mathbb{E}A_T < \infty$ for all $T > 0$. Let h be a progressively measurable process that satisfies*

$$\mathbb{E} \int_0^T h_t^2 dA_t < \infty,$$

for each $T > 0$. Then there exists a sequence $\{h^n\}$ of simple processes such that

$$\sup_{T>0} \lim_{n \rightarrow \infty} \mathbb{E} \int_0^T |h_t^n - h_t|^2 dA_t = 0.$$

PROOF: Using a standard truncation argument, we may assume w.l.o.g. that $h_t(\omega) \leq C$ for some $C > 0$ and all (t, ω) . Moreover, it suffices to construct, for each $T > 0$ a sequence $\{h^{n,T}\}$ of simple processes such that

$$\lim_{n \rightarrow \infty} \mathbb{E} \int_0^T |h_t^{n,T} - h_t|^2 dA_t = 0.$$

In this case, we can find, for each $m \in \mathbb{N}$ an integer $n_m \in \mathbb{N}$ such that

$$\mathbb{E} \int_0^m |h_t^{n_m, m} - h_t|^2 dA_t \leq \frac{1}{m}$$

and the sequence $\{h^{n_m, m}\}$ has the desired properties. In what follows we may thus fix $T > 0$ and assume that $h_t = 0$ for all $t > T$ and all $\omega \in \Omega$.

Compared to the arguments given in Proposition 4.1.5 the idea is to first change the time to replace the dA_t -integral by a dt -integral so that Proposition 4.1.5 can be applied.

More precisely, let us consider the time change $T_s(\omega)$ defined by inverse of the strictly increasing function $A_t(\omega) + t$, that is

$$A_{T_s(\omega)} + T_s(\omega) = s.$$

Then $T_s \leq s$ and $\{T_s \leq t\} = \{A_t + t \geq s\} \in \mathcal{F}_t$ so T_s is a bounded stopping time for each $s \geq 0$. Taking s as our new variable, we define the filtration

$$\mathcal{G}_s = \mathcal{F}_{T_s}$$

and introduce the measurable time-changed process

$$g_s := h_{T_s} \quad \text{hence} \quad g_{A_s + s} = h_s.$$

This process is adapted to (G_s) because of the progressive measurability of h ; see Proposition 1.2.6.

By Proposition 4.1.5 there exists, for any $\varepsilon > 0$ and $R > 0$ a (G_s) -adapted simple process g^ε such that

$$\mathbb{E} \int_0^R |g^\varepsilon - g| ds < \frac{\varepsilon}{2}.$$

Since (the first equality uses $h \equiv 0$ on $(T, \infty]$)

$$\mathbb{E} \int_0^\infty g^2 ds = \mathbb{E} \int_0^\infty \mathbf{1}_{\{T_s \leq T\}} h_{T_s}^2 ds = \mathbb{E} \int_0^{A_T + T} h_{T_s}^2 ds \leq C^2 (\mathbb{E} A_T + T) < \infty$$

by choosing R large enough and setting $g_s^\varepsilon = 0$ for $s > R$, we obtain that

$$\mathbb{E} \int_0^\infty |g_s^\varepsilon - g_s|^2 ds < \varepsilon.$$

Since g^ε is simple, we may write (assuming for simplicity that the process starts in zero)

$$g_s^\varepsilon = \sum_{j=1}^n \xi_{s_{j-1}} \mathbf{1}_{(s_{j-1}, s_j]}(s)$$

where each ξ_{s_j} is $\mathcal{F}_{T_{s_j}}$ -measurable and uniformly bounded, say by some constant $K > 0$. Reverting back to the original clock we set

$$h_t^\varepsilon := g_{A_t + t}^\varepsilon = \sum_{j=1}^n \xi_{s_{j-1}} \mathbf{1}_{(T_{s_{j-1}}, T_{s_j}]}(t).$$

By Lemma 1.2.5 vi) the random variable $\xi_{s_{j-1}} \mathbf{1}_{\{T_{s_j} \leq t\}}$ is $\mathcal{F}_t$ -measurable. Hence, h^ε is measurable and adapted and it approximates h because

$$\mathbb{E} \int_0^T |h_t^\varepsilon - h_t|^2 dA_t \leq \mathbb{E} \int_0^T |h_t^\varepsilon - h_t|^2 (dA_t + dt) \leq \mathbb{E} \int_0^T |g_t^\varepsilon - g_t|^2 dt < \varepsilon.$$

Since the process h^ε is not simple it remains to approximate the processes

$$\eta_t(\omega) := \xi_{s_{j-1}} \mathbf{1}_{(T_{s_{j-1}}, T_{s_j}]}(t)$$

that “form the basis” for h^ε by simple ones. To simplify notation let $s_{j-1} = 1$ and $s_j = 2$ so $T_1 \leq T_2 \leq 2$. Let

$$T_i^m(\omega) := \sum_{k=1}^{1+2^{m+1}} \frac{k}{2^m} \mathbf{1}_{[(k-1)/2^m, k/2^m)}(T_i(\omega)), \quad i = 1, 2$$

and define

$$\eta_t^m(\omega) := \xi_1(\omega) \mathbf{1}_{(T_1^m(\omega), T_2^m(\omega))}(t)$$

We claim that η^m is a simple process and that it approximates the process η . Indeed, noticing that the random times $T_1^m(\omega)$ and $T_2^m(\omega)$ may be equal in which case $\eta_t^m(\omega) = 0$ we can represent η^m as

$$\begin{aligned} \eta_t^m(\omega) &= \sum_{k=1}^{2^{m+1}} \xi_1(\omega) \mathbf{1}_{((k-1)/2^m, k/2^m]}(t) \mathbf{1}_{\{T_1 < (k-1)/2^m \leq T_2\}}(\omega) \\ &= \sum_{k=1}^{2^{m+1}} \xi_1(\omega) \mathbf{1}_{\{T_1 < (k-1)/2^m\}} \mathbf{1}_{((k-1)/2^m, k/2^m]}(t) \mathbf{1}_{\{T_1 < (k-1)/2^m \leq T_2\}}(\omega). \end{aligned}$$

Now,

$$\{T_1 < (k-1)/2^m \leq T_2\} \in \mathcal{F}_{(k-1)/2^m}$$

and by Lemma 1.2.5 vi),

$$\xi_1 \mathbf{1}_{\{T_1 < (k-1)/2^m\}} \in \mathcal{F}_{(k-1)/2^m}.$$

As a result, the process η^m is indeed simple. Furthermore,

$$\mathbb{E} \int_0^\infty |\eta_t^m - \eta_t|^2 dA_t \leq K^2 [\mathbb{E}(A_{T_2^m} - A_{T_2}) + \mathbb{E}(A_{T_1^m} - A_{T_1})] \rightarrow 0 \quad \text{as } m \rightarrow \infty.$$

□

Summary. So far we have shown that the stochastic integral $h \cdot M$ exists for all $M \in \mathcal{M}$ and all predictable/progressively measurable $\mathbb{P}_M$ -square integrable processes h and that $h \cdot M$ belongs to $\mathcal{M}$. Do note, though, that for a predictable process h with approximating sequence $\{h^n\}$ we have that

$$\lim_{n \rightarrow \infty} \mathbb{E} \int_0^\infty |h_t^n - h_t|^2 d\langle M \rangle_t = 0.$$

For progressively measurable (and also for predictable) processes we have that

$$\sup_{T>0} \lim_{n \rightarrow \infty} \mathbb{E} \int_0^T |h_t^n - h_t|^2 d\langle M \rangle_t = 0,$$

under the weaker requirement

$$\mathbb{E} \int_0^T h_s^2 d\langle M \rangle_s < \infty \quad \text{for all } T > 0.$$

We continue to work with predictable integrands.

4.2.3 Properties of the stochastic integral

In what follows, we establish properties for the bracket process of stochastic integrals. This requires some measure-theoretic preliminaries.

For a signed Borel measure ν on $\mathbb{R}$, we denote by $h \cdot |\nu|$ the integral of a function h w.r.t. the total variation measure $|\nu| := \nu^+ + \nu^-$ where $\nu^\pm$ denote the unique positive measures such that $\nu = \nu^+ - \nu^-$ (Hahn decomposition). We recall that

$$|h \cdot \nu| \leq |h| \cdot |\nu|.$$

Moreover, if $\nu \ll \mu$ for some positive measure μ with density $g := \frac{d\nu}{d\mu}$, then $\nu = g^+ d\mu - g^- d\mu$, and the uniqueness of the Hahn decomposition shows that

$$|\nu| = (g^+ + g^-) d\mu = |g| d\mu,$$

that is, $|\nu|$ is absolutely continuous w.r.t. μ as well with density

$$\frac{d|\nu|}{d\mu} = |g|. \quad (4.2)$$

For a function of bounded variation A with Jordan decomposition $A = A_1 - A_2$ we denote by dA the associated Lebesgue-Stieltjes measure and by $d|A|$ the total variation measure of dA .⁵

Thus, for every $M, N \in \mathcal{M}$ we can define the Lebesgue-Stieltjes integral (ω -by- ω) of h w.r.t. the total variation measure $d|\langle M, N \rangle|$, denoted

$$\int h d|\langle M, N \rangle| =: h \cdot |\langle M, N \rangle|.$$

Our first goal is to prove the Kunita-Watanabe inequality that estimates the co-variation of stochastic integrals in terms of the quadratic variation of the integrals involved. The proof requires the following result from which we shall deduce that the measure of $d\langle M_1, M_2 \rangle$ is almost surely absolutely continuous w.r.t. the measure $d\langle M_1 \rangle + \langle M_2 \rangle$.

Lemma 4.2.10 *Let μ_i , $i = 1, 2$ be finite, non-negative Borel measures on $[0, T]$ and ν be a signed Borel measure on $[0, T]$ that satisfies⁶*

$$|\nu(I)| \leq \sqrt{\mu_1(I)\mu_2(I)}$$

for any half-open interval $I \subseteq [0, T]$. Then ν and hence $|\nu|$ is absolutely continuous w.r.t. $\mu := \frac{1}{2}(\mu_1 + \mu_2)$ and the density $g = \frac{d\nu}{d\mu}$ satisfies $|g| \leq \sqrt{f_1 f_2}$ μ -a.e. where $f_i = \frac{d\mu_i}{d\mu}$ for $i = 1, 2$.

PROOF: Absolute continuity follows from the binomial formula, which gives us for all $\alpha > 0$ that

$$|\nu(I)| \leq \sqrt{\mu_1(I)\mu_2(I)} = \sqrt{\alpha\mu_1(I)\frac{1}{\alpha}\mu_2(I)} \leq \frac{1}{2}\left(\alpha\mu_1(I) + \frac{1}{\alpha}\mu_2(I)\right).$$

⁵If we denote by V the total variation function of A with associated Lebesgue-Stieltjes measure dV , then it is not difficult to show that $dV = d|A|$.

⁶We do mean the absolute value of $\nu(I)$, not $|\nu|(I)$.

In particular,

$$2 \left| \int_0^T \varphi g d\mu \right| \leq \alpha \int_0^T |\varphi| f_1 d\mu + \alpha^{-1} \int_0^T |\varphi| f_2 d\mu,$$

first for all φ that are indicator functions of intervals, then for linear combinations and finally for all bounded, Borel measurable ones. This implies that μ -almost everywhere

$$2|g| \leq \alpha f_1 + \alpha^{-1} f_2.$$

Taking the unique α^* that minimizes the right-hand side of the equation and plugging α^* back into this equations yields $|g| \leq \sqrt{f_1 f_2}$. $\square$

Corollary 4.2.11 *For $M_1, M_2 \in \mathcal{M}$, we have for each $s < t$ almost surely*

$$|\langle M_1, M_2 \rangle_t - \langle M_1, M_2 \rangle_s| \leq \sqrt{\langle M_1 \rangle_t - \langle M_1 \rangle_s} \sqrt{\langle M_2 \rangle_t - \langle M_2 \rangle_s}.$$

Equivalently, $d\langle M_1, M_2 \rangle$ is a.s. absolutely continuous w.r.t.

$$\mu_\omega := \frac{1}{2} (\langle M_1 \rangle(\omega) + \langle M_2 \rangle(\omega)).$$

If $f_{i,j}(\cdot, \omega)$ denotes the density of $d\langle M_i, M_j \rangle(\omega)$ w.r.t μ_ω , then a.s.

$$|f_{1,2}(\cdot, \omega)| \leq \sqrt{f_{1,1}(\cdot, \omega) f_{2,2}(\cdot, \omega)} \quad \mu_\omega\text{-almost everywhere.}$$

PROOF: The following (in)equality is a trivial consequence of non-negative bilinearity: for any $\alpha > 0$,

$$\begin{aligned} 0 &\leq \langle \alpha^{1/2} M_1 \pm \alpha^{-1/2} M_2, \alpha^{1/2} M_1 \pm \alpha^{-1/2} M_2 \rangle_t - \langle \alpha^{1/2} M_1 \pm \alpha^{-1/2} M_2, \alpha^{1/2} M_1 \pm \alpha^{-1/2} M_2 \rangle_s \\ &= \alpha (\langle M_1 \rangle_t - \langle M_1 \rangle_s) \pm 2 (\langle M_1, M_2 \rangle_t - \langle M_1, M_2 \rangle_s) + \alpha^{-1} (\langle M_2 \rangle_t - \langle M_2 \rangle_s). \end{aligned}$$

From this we see that

$$2 |\langle M_1, M_2 \rangle_t - \langle M_1, M_2 \rangle_s| \leq \alpha (\langle M_1 \rangle_t - \langle M_1 \rangle_s) + \frac{1}{\alpha} (\langle M_2 \rangle_t - \langle M_2 \rangle_s).$$

Minimizing the right side w.r.t. α and plugging the minimizer back into the equation yields the first assertion. The second assertion follows from the preceding lemma as

$$|d\langle M_1, M_2 \rangle|((s, t]) = |\langle M_1, M_2 \rangle_t - \langle M_1, M_2 \rangle_s|.$$

$\square$

We are now ready to state and proof the Kunita-Watanabe (in)equality.⁷

Proposition 4.2.12 *(Kunita-Watanabe identity and inequality) For $M, N \in \mathcal{M}$, $h \in \bar{L}(M)$ and $g \in \bar{L}(N)$ the following holds a.s.*

$$a) \left((hg) \cdot \langle M, N \rangle \right)_t^2 \leq (|hg| \cdot |\langle M, N \rangle|)_t^2 \leq (h^2 \cdot \langle M \rangle)_t (g^2 \cdot \langle N \rangle)_t$$

⁷Note that item b) in the following proposition follows directly from the isometry property of the stochastic integral in case $M = N$ (no matter if f and g agree or not) since an isometry between Hilbert spaces automatically preserves the inner product.

$$b) \langle h \cdot M, g \cdot N \rangle = (hg) \cdot \langle M, N \rangle.$$

$$c) \mathbb{E} [|\langle h \cdot M, g \cdot N \rangle|] \leq \sqrt{\mathbb{E} [h^2 \cdot \langle M \rangle]} \sqrt{\mathbb{E} [g^2 \cdot \langle N \rangle]}.$$

PROOF:

- a) The first inequality follows by definition. To see the second, let $d\xi := d\langle M, N \rangle$ and $d\hat{\xi} = d|\langle M, N \rangle|$. In view of Corollary 4.2.11 the measures $d\xi$ and $d\hat{\xi}$ are a.s. absolutely continuous w.r.t.

$$d\eta := \frac{1}{2} (d\langle M \rangle + d\langle N \rangle).$$

By the Radon-Nikodym theorem this implies the existence of functions $f_i(\cdot, \omega) : [0, \infty) \rightarrow \mathbb{R}$ for $i = 1, 2, 3$ such that a.s.

$$d\langle M \rangle(\omega) = f_1(\cdot, \omega) d\eta(\omega), \quad d\langle N \rangle(\omega) = f_2(\cdot, \omega) d\eta(\omega),$$

and

$$d\langle M, N \rangle(\omega) = f_3(\cdot, \omega) d\eta(\omega), \quad d|\langle M, N \rangle|(\omega) = |f_3|(\cdot, \omega) d\eta(\omega)$$

where the second equation in the last display follows from (4.2). As a result, for all $\alpha, \beta \in \mathbb{R}$ on a set $\Omega_{\alpha, \beta}$ of full measure, using the polarization identity

$$\begin{aligned} 0 &\leq \langle \alpha M + \beta N \rangle_t - \langle \alpha M + \beta N \rangle_s \\ &= \int_s^t (\alpha^2 f_1(u, \omega) + 2\alpha\beta f_3(u, \omega) + \beta^2 f_2(u, \omega)) d\eta_u(\omega); \quad 0 \leq s < t < \infty. \end{aligned}$$

Since the above inequality holds for all $0 \leq s < t < \infty$ a standard monotone class argument shows that there must exist - for every $\omega \in \Omega_{\alpha, \beta}$ - a set $T_{\alpha, \beta} \in \mathcal{B}$ with $\int_{T_{\alpha, \beta}} d\eta_t(\omega) = 0$ and

$$\alpha^2 f_1(t, \omega) + 2\alpha\beta f_3(t, \omega) + \beta^2 f_2(t, \omega) \geq 0 \quad \text{on} \quad T_{\alpha, \beta}^c.$$

This shows that there exists a set of full measure

$$\tilde{\Omega} := \bigcap_{\alpha, \beta \in \mathbb{Q}} \Omega_{\alpha, \beta}$$

and sets $T(\omega)$ such that $\int_{T(\omega)} d\eta_u(\omega) = 0$ for all $\omega \in \tilde{\Omega}$, and such that the above inequality holds for all $(\alpha, \beta) \in \mathbb{R}^2$ and all $t \notin T(\omega)$. In particular, setting

$$\alpha := a|h_t(\omega)| \quad \text{and} \quad \beta := \text{sign}(f_3(t, \omega))|g_t(\omega)|$$

we see that

$$a^2 |h_t(\omega)|^2 f_1(t, \omega) + 2a |h_t(\omega)| |g_t(\omega)| |f_3(t, \omega)| + |g_t(\omega)|^2 f_2(t, \omega) \geq 0.$$

Integrating w.r.t. $d\eta$ we obtain

$$a^2 \int_0^t |h_s|^2 d\langle M \rangle_s + 2a \int_0^t |h_s g_s| d|\langle M, N \rangle|_s + \int_0^t |g_s|^2 d\langle N \rangle_s \geq 0.$$

Minimizing over a yields the desired result.

b) Exercise

c) This is just a combination of parts a) and b) using the Cauchy-Schwarz inequality.

□

Exercise 42 Prove part b) of the preceding proposition. First, show the claim for simple integrands, then approximate using (3.4).

It follows from the above proposition that the quadratic variation of the stochastic integral $h \cdot M$ equals the integral of $|h|^2$ with respect to the quadratic variation process of M . The Kunita-Watanabe (in)equality also allows us to characterize stochastic integrals.

Proposition 4.2.13 Let $M \in \mathcal{M}$ and $h \in \bar{L}(M)$. Then the following holds.

1. The stochastic integral $h \cdot M$ is the unique martingale in $\mathcal{M}$ that satisfies for all $N \in \mathcal{M}$ that

$$\langle h \cdot M, N \rangle_t = h \cdot \langle M, N \rangle_t; \quad 0 \leq t < \infty \quad \mathbb{P}\text{-a.s.}$$

2. For any stopping time τ it holds that

$$(\mathbf{1}_{[0, \tau]} h) \cdot M = (h \cdot M)^\tau = h \cdot M^\tau.$$

PROOF:

1. We already know that the integral satisfies the desired property. Let us then assume $X \in \mathcal{M}$ is another martingale satisfying $\langle X, N \rangle = h \cdot \langle M, N \rangle$ for all $N \in \mathcal{M}$. then $\langle h \cdot M - X, N \rangle = 0$ for all $N \in \mathcal{M}$. Setting $N = h \cdot M - X$ implies that

$$\langle h \cdot M - X, h \cdot M - X \rangle \equiv 0$$

which implies that

$$\|h \cdot M - X\|_{\mathcal{M}} = \mathbb{E}[\langle h \cdot M - X \rangle_\infty] = \mathbb{E}\left[\lim_{t \rightarrow \infty} \langle h \cdot M - X \rangle_t\right] = 0.$$

That is, $h \cdot M - X$ a.s. in $\mathcal{M}$.

2. For the first equality note that

$$\langle (h \cdot M)^\tau, N \rangle = \langle h \cdot M, N \rangle^\tau \stackrel{1}{=} (h \cdot \langle M, N \rangle)^\tau = (\mathbf{1}_{[0, \tau]} h) \cdot \langle M, N \rangle \stackrel{1}{=} \langle \mathbf{1}_{[0, \tau]} h \cdot M, N \rangle.$$

For the second equality observe that

$$\langle h \cdot M^\tau, N \rangle = h \cdot \langle M^\tau, N \rangle = h \cdot \langle M, N \rangle^\tau.$$

Thus, the assertion follows from Proposition 4.2.13(1.).

□

The following corollary shows that $N = h \cdot M$ implies $g \cdot N = (gh) \cdot M$. That is, the corollary tells us how to integrate a stochastic process w.r.t. stochastic integrals (which are themselves square integrable martingales).

Corollary 4.2.14 Let $M \in \mathcal{M}$, $h \in \bar{L}(M)$ and $N = h \cdot M$. If $g \in \bar{L}(N)$, then $hg \in \bar{L}(M)$ and $g \cdot N = (hg) \cdot M$.

PROOF: Using that $d\langle N \rangle_t = h_t^2 d\langle M \rangle_t$ we have that

$$\mathbb{E} \int_0^\infty h_t^2 g_t^2 d\langle M \rangle_t = \mathbb{E} \int_0^\infty g_t^2 d\langle N \rangle_t < \infty$$

and so $hg \in \overline{L}(M)$. For any $\tilde{N} \in \mathcal{M}$ the cross-variation formula for stochastic integrals yields

$$d\langle N, \tilde{N} \rangle = h d\langle M, \tilde{N} \rangle$$

and so

$$\langle (hg) \cdot M, \tilde{N} \rangle_t = \int_0^t h_s g_s d\langle M, \tilde{N} \rangle_s = \int_0^t g_s d\langle N, \tilde{N} \rangle_s = \langle g \cdot N, \tilde{N} \rangle_t.$$

□

4.2.4 Stochastic integration w.r.t. continuous local martingales

We are now going to extend our definition of stochastic integrals to continuous local martingales for integrands from the following set.

Definition 4.2.15 For a continuous local martingale $M \in \mathcal{M}_{loc}^0$ starting in zero, let

$$\mathcal{L}_{loc}(M) := \left\{ h \text{ progressive} : \int_0^t h_s^2 d\langle M \rangle_s < \infty \text{ a.s. for all } t \in (0, \infty) \right\}.$$

Remark 4.2.16 If the quadratic variation process of M is a.s. absolutely continuous as it is the case for Brownian motion, then progressive measurability is not required; it is enough to assume that h is measurable and adapted.

We are going to define the following localizing sequences of stopping times. For $M \in \mathcal{M}_{loc}^0$ and $h \in \mathcal{L}_{loc}(M)$ let $\{\tau_n\}$ be a sequence of stopping times such that $\tau_n \uparrow \infty$ and $M^{\tau_n} \in \mathcal{M}$ for all $n \in \mathbb{N}$. Further, let

$$\nu_n := n \wedge \inf \left\{ t \geq 0 : \int_0^t |h_s|^2 d\langle M \rangle_s \geq n \right\}$$

and let $\sigma_n := \tau_n \wedge \nu_n$. Then, $\sigma_n \rightarrow \infty$ a.s. and

$$M^n := M^{\sigma_n} \in \mathcal{M}, \quad h_t^n := h_t \mathbf{1}_{\{t \leq \sigma_n\}} \in \overline{L}(M^n).$$

Definition 4.2.17 For $M \in \mathcal{M}_{loc}^0$ and $h \in \mathcal{L}_{loc}(M)$ let

$$(h \cdot M)_t := \int_0^t h_s dM_s := (h^n \cdot M^n)_t \quad \text{on } \{t \leq \sigma^n\}$$

We need to make sure that the above definition makes sense. In other words, we need to show that stochastic integrals are determined locally by the local values of the integrator and integrand. This is achieved by the following corollary. The proof is left as an exercise.

Corollary 4.2.18 et $M, N \in \mathcal{M}$ and $h \in \overline{L}(M)$, $g \in \overline{L}(N)$ and let τ be a stopping time. If, for $\mathbb{P}$ -a.e. ω ,

$$h_{t \wedge \tau}(\omega) = g_{t \wedge \tau}(\omega); \quad M_{t \wedge \tau}(\omega) = N_{t \wedge \tau}(\omega) \quad \text{for all } t \geq 0,$$

then

$$(h \cdot M)(\omega)_{t \wedge \tau} = (g \cdot N)(\omega)_{t \wedge \tau} \quad \text{for all } t \geq 0.$$

Exercise 43 *Proof the above corollary.*

Roughly speaking, all properties derived for stochastic integrals in case $M \in \mathcal{M}$ and $h \in \overline{\mathcal{L}}(M)$ which do not involve expected values carry over to the general case (and are easily proved). Those properties which do involve expected values will generally not carry over. For example, stochastic integrals w.r.t. local martingales cannot be expected to be martingales; in general they are only local martingales.

We close this section with two convergence results for stochastic integrals. This first characterizes the stochastic integral $h \cdot M$ in terms of a Lebesgue-Stieltjes integral.

Proposition 4.2.19 *Let $M \in \mathcal{M}_{loc}^0$ and h be adapted and continuous (hence predictable). Let $\Pi^n = \{t_i^n : i \in \mathbb{N}\}$ be a partition of $\mathbb{R}_+$ with $|\Pi^n| \rightarrow 0$. Then*

$$\mathbb{P} - \lim_{n \rightarrow \infty} \sup_{s \leq t} \left| \sum_{k=0}^{\infty} h_{t_k^n} (M_{t_{k+1}^n \wedge s} - M_{t_k^n \wedge s}) - \int_0^s h_r dM_r \right| = 0 \quad \text{for all } t \in [0, \infty).$$

PROOF: Let

$$h_s^n := \sum_{k=0}^{\infty} h_{t_k^n} \mathbf{1}_{(t_k^n, t_{k+1}^n]}(s).$$

Continuity of h allows us to choose a localizing sequence (τ_m) such that $h \mathbf{1}_{[0, \tau_m]}$ is uniformly bounded in (t, ω) , for all $m \in \mathbb{N}$. In particular, $h^n \mathbf{1}_{[0, \tau_m]}$ is a simple process for all $n, m \in \mathbb{N}$ and

$$h \mathbf{1}_{[0, \tau_m]} \in \mathcal{L}(M^m)$$

for all $m \in \mathbb{N}$ where $M^m := M^{\tau_m}$. Since $t \mapsto h \mathbf{1}_{[0, \tau_m]}$ is still adapted and continuous, hence predictable, it follows as in the proof of Proposition 4.2.8 that

$$\|h^n \mathbf{1}_{[0, \tau_m]} - h \mathbf{1}_{[0, \tau_m]}\|_{M^m} \rightarrow 0 \quad \text{as } n \rightarrow \infty$$

and hence the corresponding stochastic integrals w.r.t. M^m converge ucp for all $m \in \mathbb{N}$. The result now follows from the definition of stochastic integrals w.r.t. local martingales. $\square$

We emphasize that we did not require $h \in \mathcal{L}_{loc}(M)$ that is, integrability of h , in the above proposition. This assumption was “replaced” by stopping the integrands; they were bounded after stopping and the stopped integrands were still predictable because they were still adapted and continuous. In particular, we have the following corollary.

Corollary 4.2.20 *If h is adapted and continuous, then the stochastic integral $h \cdot M$ exists for all continuous local martingales M .*

The following “dominated convergence-type” result for stochastic integrals can be viewed as a generalization of Proposition 4.2.19.

Lemma 4.2.21 *Let $M \in \mathcal{M}_{loc}^0$ and $h, h^n \in \mathcal{L}_{loc}(M)$ for all $n \in \mathbb{N}$. Suppose that $h^n \rightarrow h$ uniformly in probability on compact sets (ucp). Then*

$$h^n \cdot M \rightarrow h \cdot M \quad \text{ucp.}$$

PROOF: We may w.l.o.g. assume that $h = 0$. Let

$$\tau_m := \inf \{s \geq 0 : |M_s| \geq m\}; \quad M_s^m := M_{s \wedge \tau_m}; \quad \tau_{m,n} := \inf \left\{ s \geq 0 : \int_0^s |h_u^n|^2 d\langle M^m \rangle_u \geq \frac{1}{m} \right\}.$$

Then, it holds for all $t, \delta > 0$ that (recall that $h = 0$)

$$\mathbb{P} \left(\sup_{0 \leq s \leq t} \left| \int_0^s h_u^n dM_u \right| \geq \delta \right) \leq \mathbb{P}(\tau_{m,n} \wedge \tau_m \leq t) + \mathbb{P} \left(\sup_{0 \leq s \leq t} \left| \int_0^{s \wedge \tau_{m,n} \wedge \tau_m} h_u^n dM_u \right| \geq \delta \right).$$

Using first Chebychev's inequality, then Doob's L^2 -maximal inequality and then that fact that $\|M\|_{\mathcal{M}}^2 = \mathbb{E}[\langle M \rangle_\infty]$ for any $M \in \mathcal{M}$ the last term can be estimated from above by

$$\mathbb{P} \left(\sup_{0 \leq s \leq t} \left| \int_0^s h_u^n \mathbf{1}_{[0, \tau_{m,n}]} dM_u^m \right| \geq \delta \right) \leq \frac{4}{\delta^2} \mathbb{E} \left[\int_0^t (h_u^n \mathbf{1}_{[0, \tau_{m,n}]})^2 d\langle M^m \rangle_u \right] \leq \frac{4}{\delta^2 m}.$$

We notice that Doob's inequality applies because the stochastic integral is a proper martingale, due to the stopped integrand.

For given $\varepsilon, \delta > 0$, we now first choose $m \in \mathbb{N}$ such that

$$\mathbb{P}(\tau_m \leq t) \leq \varepsilon/3 \quad \text{and} \quad \frac{4}{\delta^2 m} \leq \varepsilon/3.$$

Having fixed a martingale M^m we now use the uniform ucp-convergence of the sequence (h^n) to deduce from

$$\int_0^s |h_u^n|^2 d\langle M^m \rangle_u \leq \left(\sup_{0 \leq u \leq s} |h_u^n| \right) \langle M^m \rangle_s \xrightarrow{\mathbb{P}} 0$$

that we can choose $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$ it holds that $\mathbb{P}(\tau_{m,n} \leq t) \leq \varepsilon/3$. For those $n \in \mathbb{N}$ we then obtain that

$$\mathbb{P} \left(\sup_{0 \leq s \leq t} \left| \int_0^s h_u^n dM_u \right| \geq \delta \right) \leq \varepsilon.$$

□

4.3 The Itô formula

Our next aim is to formulate and prove Itô's formula, the chain rule of stochastic calculus. An adapted process X is a semimartingale if it allows the decomposition

$$X_t = x + M_t + A_t$$

where M is a continuous local martingale starting in zero and A is a process of bounded variation, also starting in zero. By Lemma 3.3.10 this decomposition is unique. We start with the integration by parts formula.

Theorem 4.3.1 *Let X, Y be continuous semi-martingales. Then,*

$$X_t Y_t - X_0 Y_0 = \int_0^t X_s dY_s + \int_0^t Y_s dX_s + \langle X, Y \rangle_t.$$

PROOF: Let $\Pi^n = \{t_i^n : i \in \mathbb{N}\}$ be a partition of $[0, \infty)$. Then

$$\begin{aligned} X_t Y_t - X_0 Y_0 &= \sum_{k=0}^{\infty} (X_{t_{k+1}^n \wedge t} - X_{t_k^n \wedge t}) (Y_{t_{k+1}^n \wedge t} - Y_{t_k^n \wedge t}) \\ &\quad + \sum_{k=0}^{\infty} Y_{t_k^n \wedge t} (X_{t_{k+1}^n \wedge t} - X_{t_k^n \wedge t}) + \sum_{k=0}^{\infty} X_{t_k^n \wedge t} (Y_{t_{k+1}^n \wedge t} - Y_{t_k^n \wedge t}). \end{aligned}$$

The convergence of the last two sums to the corresponding integrals follows from our Riemann-type approximation result for stochastic integrals. The convergence of the first term follows by polarization and the previously established fact that the sum of quadratic increments of continuous local martingales converges in probability uniformly on compact time intervals to the bracket process. $\square$

We are now ready to state and prove the Itô formula for continuous semi-martingales. The proof uses the fact that the class of polynomials is dense in the set of continuous functions with compact supports (Weierstrass' approximation result).

Theorem 4.3.2 (*Itô formula*) Let $f \in C^2(\mathbb{R}^d)$ and let $X = (X^1, \dots, X^d)$ be a vector of continuous semi-martingales. Then, a.s.

$$f(X_t) = f(X_0) + \sum_{i=1}^d \int_0^t \frac{\partial f}{\partial x_i}(X_u) dX_u^i + \frac{1}{2} \sum_{i,j=1}^d \frac{\partial^2 f}{\partial x_i \partial x_j}(X_u) d\langle M^i, M^j \rangle_u$$

where $dX^i = dM^i + dA^i$ and

$$\langle M^i, M^j \rangle := \frac{1}{4} [\langle M^i + M^j \rangle + \langle M^i - M^j \rangle].$$

PROOF: Obviously, Itô's formula holds for constant F as well as for $F(x) = x_k$ for some $k \in \{1, \dots, d\}$. Moreover, it is straightforward to check that Itô's formula for $F(x) = x_k x_l$ for some $k, l \in \{1, \dots, d\}$ is the same as the product rule that we established above.

Further, if the formula holds for $F, G \in C^2(\mathbb{R}^d)$, then it holds for each linear combination of F and G , so we already know that the formula is true for all polynomials of order at most 2. In a first step we are now going to extend the formula to general polynomials; then we use Weierstrass' approximation theorem to conclude.

- If the formula holds for $F \in C^2$, then it holds for $G(x) = x_i F(x)$. Indeed, by first applying the product rule to G and then Itô's formula to $F(X)$ we get that

$$\begin{aligned} G(X_t) - G(X_0) &= X_t^i F(X_t) - X_0^i F(X_0) \\ &= \int_0^t X_s^i dF(X_s) + \int_0^t F(X_s) dX_s^i + \langle X^i, F(X) \rangle_t \\ &= \int_0^t \sum_k X_s^i \frac{\partial F(X_s)}{\partial x_k} dX_s^k + \frac{1}{2} \int_0^t \sum_{k,l} X_s^i \frac{\partial^2 F(X_s)}{\partial x_k \partial x_l} d\langle X^k, X^l \rangle_s \\ &\quad + \int_0^t F(X_s) dX_s^i + \langle X^i, F(X) \rangle_t. \end{aligned}$$

Applying Itô's formula to again to $F(X)$ and recalling that the cross-variation of two processes vanishes if one process is of bounded variation yields that

$$\langle X^i, F(X) \rangle_t = \sum_k \int_0^t \frac{\partial F}{\partial x_k}(X_s) d\langle X^i, X^k \rangle_s.$$

Moreover,

$$\frac{\partial G}{\partial x_k} = x_i \frac{\partial F}{\partial x_k} + \delta_{i,k} F, \quad \frac{\partial^2 G}{\partial x_k \partial x_l} = x_i \frac{\partial^2 F}{\partial x_k \partial x_l} + \delta_{i,k} \frac{\partial F}{\partial x_k} + \delta_{i,l} \frac{\partial F}{\partial x_l}.$$

We conclude that Itô's formula holds for G and hence for all polynomials.

- By Weierstrass' approximation result F along with its first two derivatives can be uniformly approximated by polynomials p^n on compact sets. Thus, if X almost never leaves some compact set $K := [-k, k]^d$, then it follows from the preceding lemma that

$$\begin{aligned} p^n(X_t) - F(X_0) &= \sum_{i=1}^d \int_0^t \frac{\partial p^n}{\partial x_i}(X_s) dX_s^i + \frac{1}{2} \sum_{i,j=1}^d \int_0^t \frac{\partial^2 p^n}{\partial x_i \partial x_j}(X_s) d\langle X^i, X^j \rangle_s \\ &\xrightarrow{n \rightarrow \infty} \sum_{i=1}^d \int_0^t \frac{\partial F}{\partial x_i}(X_s) dX_s^i + \frac{1}{2} \sum_{i,j=1}^d \int_0^t \frac{\partial^2 F}{\partial x_i \partial x_j}(X_s) d\langle X^i, X^j \rangle_s \end{aligned}$$

where the converge holds ucp, due to Lemma 4.2.21 and

$$p^n(X_t) \xrightarrow{n \rightarrow \infty} F(X_t).$$

- For a general continuous semimartingale X we may w.l.o.g. assume that $x = 0$ and define the sequence of stopping times

$$\tau^k := \inf\{t \geq 0 \mid X_t \notin [-k, +k]^d\}.$$

The stopped process $X^k := X^{\tau^k}$ is a bounded semimartingale. The Itô formula holds for all the stopped process; thus, there exist sets of full measure Ω_k such that for every $k \in \mathbb{N}$ and all $\omega \in \Omega_k$ the equation holds on $[0, \tau^k(\omega)]$.

The set $\Omega^* := \bigcap_k \Omega_k$ has full measure. Since $\tau^k \uparrow \infty$ the equation actually holds on $\bigcup_k [0, \tau^k(\omega)] = [0, \infty)$ for all $\omega \in \Omega^*$. More precisely, by Lemma 4.2.21 it holds for all $t \geq 0$ that

$$\begin{aligned} F(X_t^k) - F(X_0^k) &= \sum_{i=1}^d \int_0^t \frac{\partial F}{\partial x_i}(X_s^k) dX_s^{k,i} + \frac{1}{2} \sum_{i,j=1}^d \int_0^t \frac{\partial^2 F}{\partial x_i \partial x_j}(X_s) d\langle X^i, X^j \rangle_s \\ &= \sum_{i=1}^d \int_0^{t \wedge \tau^k} \frac{\partial F}{\partial x_i}(X_s) dX_s^i + \frac{1}{2} \sum_{i,j=1}^d \int_0^{t \wedge \tau^k} \frac{\partial^2 F}{\partial x_i \partial x_j}(X_s) d\langle X^i, X^j \rangle_s \\ &\xrightarrow[k \rightarrow \infty]{ucp} \sum_{i=1}^d \int_0^t \frac{\partial F}{\partial x_i}(X_s) dX_s^i + \frac{1}{2} \sum_{i,j=1}^d \int_0^t \frac{\partial^2 F}{\partial x_i \partial x_j}(X_s) d\langle X^i, X^j \rangle_s. \end{aligned}$$

□

Exercise 44 Let (B_t) and (W_t) be two independent one dimensional Brownian motions. Find dX_t for each of the following cases:

(i) $X_t = B_t \exp(W_t + t),$

(ii) $X_t = \sin(B_t + \int_0^t W_s ds)$

4.3.1 Examples and applications

In this section we consider some examples in a Brownian setting including exponential martingales and probabilistic representations of boundary value problems. In particular, we show that the 2-dimensional Brownian motion is recurrent while the 3-dimensional Brownian motion is transient. The latter allows us to construct a local martingale that is not a martingale.

We start with Burkholder-Davis-Gundy (BDG) inequality. This inequality relates the maximum of a local martingale with its quadratic variation.

Theorem 4.3.3 (BDG inequality) *For any $p \geq 2$ there exists a constant C_p such that for any $M \in \mathcal{M}_{loc}^0$ and any $t \geq 0$ the following holds:*

$$\mathbb{E} \left[\sup_{0 \leq s \leq t} |M_s|^p \right] \leq C_p \mathbb{E} \left[\langle M \rangle_t^{p/2} \right].$$

PROOF: By stopping it suffices to consider the case where M and its quadratic variation are bounded. Since the function $m \mapsto |m|^p$ is smooth for $p \geq 2$ it follows by Itô's formula that

$$|M_t|^p = p \int_0^t \text{sign}(M_s) |M_s|^{p-1} dM_s + \frac{p(p-1)}{2} \int_0^t |M_s|^{p-2} d\langle M \rangle_s$$

Since the stochastic integral is a true martingale starting in zero, it drops out when taking expected values. Thus, by Doob's maximal inequality there exists a constant $C > 0$ such that (recall that $M_t^* := \sup_{s \leq t} |M_s|$)

$$\begin{aligned} \mathbb{E} |M_t^*|^p &\leq C \mathbb{E} \left[\int_0^t |M_s|^{p-2} d\langle M \rangle_s \right] \\ &\leq C \mathbb{E} [|M_t^*|^{p-2} \langle M \rangle_t] \\ &\leq C (\mathbb{E} [|M_t^*|^p])^{1-2/p} \left(\mathbb{E} [\langle M \rangle_t^{p/2}] \right)^{2/p} \quad (\text{Hölder with } p' = \frac{p}{p-2}; q' = \frac{p}{2}) \end{aligned}$$

Dividing both sides by $(\mathbb{E} [|M_t^*|^p])^{1-2/p}$ and then raising both sides to the power $p/2$ yields the result. $\square$

Remark 4.3.4 *It follows from the BDG inequality that a local martingale M is a martingale if $\mathbb{E}[\langle M \rangle_\infty] < \infty$; in this case M^* and hence M is L^2 -bounded.*

Next, we introduce exponential martingales. Such martingales will be very important below when we introduce equivalent changes of measures and Girsanov's theorem.

Example 4.3.5 (Exponential martingale) *Let $L \in C[0, \infty)$ and $X_t = \int_0^t L_u dB_u$. The process*

$$\mathcal{E}(X)_t := e^{X_t - \frac{1}{2} \int_0^t L_u^2 du} = e^{X_t - \frac{1}{2} \langle X \rangle_t}$$

is called exponential martingale. Applying Itô's formula to $f(t, x) = e^{x - \frac{1}{2} \int_0^t L_u^2 du}$ gives, a.s.

$$\mathcal{E}(X)_t = 1 + \int_0^t f(u, X_u) L_u dB_u.$$

We now check that $\mathcal{E}(X)$ is indeed a martingale. We have that

$$\int_0^t \mathbb{E}|f(u, X_u)L_u|^2 du \leq \sup_{0 \leq u \leq t} L_u^2 \int_0^t \mathbb{E}e^{2X_u} du. \quad (4.3)$$

Hence, we need to compute $\mathbb{E}e^{2X_u}$. Since $X_u \sim N(0, \sigma^2(u))$ with $\sigma^2(u) = \int_0^u L_s^2 ds$ (check this as an exercise using the L^2 approximation of the stochastic integral; it is important that L is a deterministic function) we get that $\mathbb{E}e^{2X_u} = e^{2\sigma^2(u)}$. In view of (4.3) this shows that $f(\cdot, X)L \in \mathcal{L}(B)$ and hence that

$$(\mathcal{E}(X)_t - 1)_{t \geq 0} \in \mathcal{M}.$$

In differential form the process $\mathcal{E}(X)$ can be expressed as

$$d\mathcal{E}(X)_t = \mathcal{E}(X)_t dX_t.$$

Since this is similar to the exponential function, $\mathcal{E}(X)$ is also called the stochastic exponential of X .

We can write down the stochastic exponential for much more general local martingales. The verification of the martingale property relied heavily on the deterministic integrand, though. We shall consider more general situations in Chapter 5.

Another important application of Itô's formula is the link between Brownian motion and the Laplace operator. Specifically, Itô's formula allows for probabilistic representations of solutions to second order linear PDEs.

Example 4.3.6 (Probabilistic representation of boundary value problems) Suppose that the function

$$F \in C_b([0, T] \times \bar{\mathcal{D}}) \cap C^{1,2}((0, T) \times \mathcal{D})$$

solves the reverse heat equation

$$F_t(t, x) + \frac{1}{2} F_{xx}(t, x) = 0$$

on the bounded domain $\mathcal{D} \subset \mathbb{R}$ with bounded boundary condition h on $(T \times \mathcal{D}) \cup ((0, T) \times \partial\mathcal{D})$. Let B be a one-dimensional standard Brownian motion and let

$$\tau := \inf\{t : B_t \notin \mathcal{D}\} \wedge T.$$

By Itô's formula, we have that

$$F(\tau, B_\tau) = F(0, 0) + \int_0^\tau F_t(s, B_s) ds + \int_0^\tau F_x(s, B_s) dB_s + \frac{1}{2} \int_0^\tau F_{xx}(s, B_s) ds.$$

The stochastic integral is a (true) martingale on $[0, \tau]$ because $\mathcal{D}$ is bounded. Thus, $\mathbb{E} \int_0^\tau F_x(s, B_s) dB_s = 0$. Since F satisfies the reverse heat equation,

$$F(\tau, B_\tau) = F(0, 0) + \int_0^\tau F_{xx}(s, B_s) ds,$$

and since $F(\tau, B_\tau) = h(\tau, B_\tau)$ we conclude that

$$F(0, 0) = \mathbb{E}h(\tau, B_\tau).$$

Similarly, we can start the Brownian motion at time t in x . If we denote by $\mathbb{P}_{t,x}$ its resulting distribution and by $\mathbb{E}_{t,x}$ the expected value with respect to $\mathbb{P}_{t,x}$, then

$$F(t, x) = \mathbb{E}_{t,x} h(\tau, B_\tau).$$

Hence we obtain a probabilistic representation of the solution to the heat equation with boundary condition h .

Next, we study transience and recurrence properties of multi-dimensional Brownian motion. As a byproduct we shall construct a local martingale that is not a martingale.

We use the following facts about Brownian motion that have been proved in Stochastic II. If W is a standard one-dimensional Brownian motion and W^* its running maximum, then

$$\mathbb{P}[W_t^* > a] = \mathbb{P}[|W_t| > a].$$

In particular, for any positive numbers $a, b > 0$ the stopping times

$$\tau_a := \inf\{t \geq 0 : W_t > a\}, \quad \tau_b := \inf\{t \geq 0 : W_t < -b\}, \quad \tau := \tau_a \wedge \tau_b.$$

are almost surely finite and $\mathbb{P}[\tau_a < \tau_b] = \frac{b}{a+b}$ and $\mathbb{P}[\tau_b < \tau_a] = \frac{a}{a+b}$.

Example 4.3.7 (2-dimensional Brownian motion is recurrent) Let $D \subset \mathbb{R}^d$ be open. A function $f \in C^2(D)$ is called harmonic if $\Delta f := \sum_{i=1}^d f_{x_i x_i} = 0$. In this case, if $0 \in D$ and B is a d -dimensional Brownian motion, it follows from Itô's formula that

$$f(B_t) = f(0) + \int_0^t \nabla f(B_s) dB_s$$

as long as B does not leave D . Let us now consider the special case where $f : \mathbb{R}^2 \setminus 0 \rightarrow \mathbb{R}$ be given by

$$f(x) = \ln |x|;$$

here, $|x| := \sqrt{\sum_{i=1}^2 x_i^2}$ denotes the Euclidean norm. Then,

$$f_{x_i} = \frac{x_i}{|x|^2}, \quad f_{x_i x_i} = \frac{1}{|x|^2} - \frac{2x_i^2}{|x|^4}.$$

Thus, f is harmonic on $\mathbb{R}^2 \setminus 0$ and hence also on the open disc $D_{r,R} := \{x \in \mathbb{R}^2 : r < |x| < R\}$. We now show that if we start a Brownian motion at some point in the plane, then it will return to any neighborhood of the origin - and hence any other point - infinitely often. We shall see that this is not the case for 3-dimensional Brownian motions. Let

$$h(x) = \frac{\ln R - \ln |x|}{\ln R - \ln r}.$$

Then h is harmonic on $D_{r,R}$ and $h(x) = 0$ if $|x| = R$ and $h(x) = 1$ if $|x| = r$. Let $x \in D_{r,R}$ and

$$\mathbb{P}_x := \mathbb{P} \circ (x + B)^{-1}$$

be the law of a Brownian motion starting in x . Let

$$\tau_r := \inf\{t \geq 0 : |B_t + x| = r\}, \quad \tau_R := \inf\{t \geq 0 : |B_t + x| = R\}, \quad \tau := \tau_r \wedge \tau_R.$$

Then τ is a stopping time and $\mathbb{P}_x(\tau < \infty) = 1$. Moreover, $h(B)$ and $\nabla h(B)$ are $\mathbb{P}_x$ -a.s. bounded on $[0, \tau]$. Thus, by Itô's formula

$$h(x) = \mathbb{E}_x h(B_\tau) = \mathbb{P}_x(\tau_r < \tau_R).$$

In view of the previously recalled results on Brownian exit times, $\tau_R \rightarrow \infty$ a.s. for $R \rightarrow \infty$. Since $h(x) \rightarrow 1$ if $R \rightarrow \infty$ for all $|x| > r > 0$ it follows from dominated convergence that $\mathbb{P}_x(\tau_r < \infty) = 1$. That is, if we start a 2-dimensional Brownian motion at some point $x \neq 0$, then it will return to any neighbourhood of the origin infinitely often.

Exercise 45 (3-dimensional Brownian motion is transient) Show that the Brownian motion in dimension $d \geq 3$ is transient and that the sequence $\{\tau_{n^{-1}}\}$ converges to $+\infty$ a.s. as $n \rightarrow \infty$.

Hint. Proceed as in the preceding example but with

$$h(x) = \frac{R^{2-d} - |x|^{2-d}}{\ln R^{2-d} - r^{2-d}}.$$

Example 4.3.8 (A local martingale that is not a martingale) Related to the previous example/exercise we are now going to consider a local martingale that is not a strict martingale. To this end, let B be a 3-dimensional Brownian motion, let $f(x) = |x|^{-1}$ for $x \in \mathbb{R}^3 \setminus \{0\}$, and let τ_r be the stopping time introduced in the previous examples. Since f is harmonic it follows by Itô's formula that

$$f(x + B_{t \wedge \tau_r}) = x + \int_0^{t \wedge \tau_r} \nabla f(x + B_s) dB_s.$$

Hence the process $R_t = f(x + B_t)$ is a bounded martingale on $[0, \tau_r]$. Moreover, it follows from Exercise 45 that the sequence $\{\tau_{n^{-1}}\}$ defines a localising sequence. Thus, (R_t) is a local martingale. If the process (R_t) were a (true) martingale, then it would have constant expectation. To see that this is not the case, we first recall that the sum of the squares of n independent $N(0, 1)$ distributed random variables is χ_n^2 -distributed. More generally, the sum X of n independent squared $N(\mu_i, 1)$ distributed random variables follows an $\chi_n^2(\lambda)$ distribution (non-central chi-square distribution) where $\lambda = \sum_i \mu_i^2$ and $\mathbb{E}X = n + \lambda$. Since

$$\frac{1}{\sqrt{t}} (B_t^i + x^i) \sim N(x^i/\sqrt{t}, 1), \quad i = 1, 2, 3$$

we see that the expected value of $R_t^{-2} = \sum_{i=1}^3 (B_t^i + x^i)^2$ is given by $\mathbb{E}R_t^{-2} = 3t + |x|^2$. By Jensen's inequality $(\mathbb{E}R_t)^{-2} \leq \mathbb{E}(R_t^{-2})$ and so

$$\mathbb{E}R_t \leq \frac{1}{\sqrt{3t + |x|^2}} \rightarrow 0 \quad \text{as } t \rightarrow \infty.$$

If (R_t) were a martingale, then this would contradict $\mathbb{E}R_0 = \frac{1}{|x|} > 0$.

4.3.2 The Feynman-Kac formula

In the previous subsection we illustrated how the solution to the heat equation with given terminal or boundary values can be represented stochastically. In this section we extend the result to a more general class of linear PDEs. Specifically, we consider the following Cauchy problem:

$$\begin{aligned} -\frac{\partial v}{\partial t} + kv &= \frac{1}{2}\Delta v + g \quad \text{on } [0, T) \times \mathbb{R}^d \\ v(T, x) &= f(x) \quad \text{on } \mathbb{R}^d \end{aligned} \tag{4.4}$$

where $f : \mathbb{R}^d \rightarrow \mathbb{R}, k : \mathbb{R}^d \rightarrow [0, \infty), g : [0, T] \times \mathbb{R}^d \rightarrow [0, \infty)$ are continuous.

Theorem 4.3.9 (Feynman-Kac formula) Suppose that $v \in C^{1,2}$ satisfies the Cauchy problem (4.4) and assume that

$$\max_{0 \leq t \leq T} |v(t, x)| + \max_{0 \leq t \leq T} |g(t, x)| \leq K e^{a|x|^2}$$

for some $K > 0$ and $a < \frac{1}{2dT}$. Then it admits the representation

$$\begin{aligned} v(t, x) = & \mathbb{E}_x \left[f(B_{T-t}) \exp \left(- \int_0^{T-t} k(B_s) ds \right) \right. \\ & \left. + \int_0^{T-t} g(t + \theta, B_\theta) \exp \left(- \int_0^\theta k(B_s) ds \right) d\theta \right]; \quad \text{on } [0, T] \times \mathbb{R}^d. \end{aligned}$$

in terms of a d -dimensional Brownian motion $B = (B^1, \dots, B^d)$ starting in x .

PROOF: Using the fact that v satisfies our Cauchy problem and that the exponential term is of bounded variation we obtain from Itô's formula (for fixed $t \in [0, T]$) that

$$\begin{aligned} & d \left[v(t + \theta, B_\theta) \exp \left(- \int_0^\theta k(B_s) ds \right) \right] \\ &= \exp \left(- \int_0^\theta k(B_s) ds \right) \left[-g(t + \theta, B_\theta) d\theta + \sum_{i=1}^d \frac{\partial}{\partial x_i} v(t + \theta, B_\theta) dB_\theta^i \right]. \end{aligned}$$

Let $\tau_n := \inf\{t \geq 0 : |B_t| \geq n\sqrt{d}\}$ and $0 < r < T - t$ and integrate the above equation on $[0, r \wedge \tau_n]$. The resulting stochastic integrals are martingales so

$$\begin{aligned} v(t, x) = & \mathbb{E}_x \int_0^{r \wedge \tau_n} g(t + \theta, B_\theta) \exp \left(- \int_0^\theta k(B_s) ds \right) d\theta \\ & + \mathbb{E}_x \left[v(t + \tau_n, B_{\tau_n}) \exp \left(- \int_0^{\tau_n} k(B_s) ds \right) \mathbf{1}_{\{\tau_n \leq r\}} \right] \\ & + \mathbb{E}_x \left[v(t + r, B_r) \exp \left(- \int_0^r k(B_s) ds \right) \mathbf{1}_{\{\tau_n > r\}} \right]. \end{aligned}$$

The first term on the right-hand side converges to

$$\mathbb{E}_x \int_0^{T-t} g(t + \theta, B_\theta) \exp \left(- \int_0^\theta k(B_s) ds \right) d\theta \quad (4.5)$$

as $n \rightarrow \infty$ and $r \rightarrow T - t$ by monotone convergence because of the non-negativity of k and the exponential growth condition on g which guarantees that it has finite expectation.^[8]

Using the growth condition on v the second term is dominated by

$$K e^{adn^2} \mathbb{P}_x[\tau_n \leq T] \leq K e^{adn^2} \sum_{j=1}^d \mathbb{P}_x[\max_{0 \leq t \leq T} |B^j| \geq n].$$

By Lemma 2.5.2 we know that $\int_x^\infty e^{-y^2/2} dy \leq \frac{1}{x} e^{-x^2/2}$. Using the fact that the running maximum W^* of a standard Brownian motion W satisfies $\mathbb{P}(W_t^* \geq x) = 2\mathbb{P}(W_t \geq x)$ we conclude that (shift by the

⁸Note that $\frac{1}{T}|B_T|^2 \sim \chi_d^2$ and recall that the moment generating function of a χ_n^2 -distributed random variable X is given by $\mathbb{E}[e^{tX}] = \frac{1}{(1-2t)^{n/2}}$ for $t < \frac{1}{2}$.

starting value, rescale by time)

$$\mathbb{P}_x[\max_{0 \leq t \leq T} |B^j| \geq n] \leq 2\sqrt{\frac{T}{2\pi}} \frac{1}{n} e^{-(n/x)^2/2T}.$$

Since $0 < a < \frac{1}{2dT}$ the second term converges to 0 as $n \rightarrow \infty$. The third term converges again by dominated convergence to

$$\mathbb{E}_x \left[v(T-t, (B_{T-t})) \exp \left(- \int_0^{T-t} k(B_s) ds \right) \right] = \mathbb{E}_x \left[f(B_{T-t}) \exp \left(- \int_0^{T-t} k(B_s) ds \right) \right]$$

as $n \rightarrow \infty$ and $r \rightarrow T-t$. □

Remark 4.3.10 A sufficient condition for the existence of a $C^{1,2}$ solution is that k is bounded and uniformly Hölder continuous, that g is Hölder continuous in the space variable uniformly in (t, x) and that g and f are of polynomial growth:

$$\max_{1 \leq t \leq T} |g(t, x)| + |f(x)| \leq L(1 + \|x\|^\nu), \quad \text{for some } \nu > 0.$$

The one-dimensional Cauchy problem (4.4) with $g = 0$ can be reduced to the heat equation as shown the following exercise.

Exercise 46 Let u satisfy the one-dimensional heat equation

$$\begin{aligned} u_t &= \frac{1}{2} u_{xx} \\ u(0, x) &= \psi(x). \end{aligned}$$

i) Solve the initial value problem

$$\begin{aligned} v_t &= \left(\frac{1}{2} \beta^2 - \alpha \right) v + \beta v_x + \frac{1}{2} v_{xx} \\ v(0, x) &= e^{-\beta x} \psi(x). \end{aligned} \tag{4.6}$$

Hint. The solution can be expressed in terms of the solution to the heat equation.

ii) Use part (i) to solve, for $a > 0$ the initial value problem

$$\begin{aligned} w_t &= \frac{a}{2} w_{xx} + b w_x + c w \\ v(0, x) &= \Psi(x). \end{aligned} \tag{4.7}$$

Hence, any solution to (4.7) can be traced back to a solution to the heat equation. Uniqueness of solutions to the heat equation thus yields uniqueness of solutions to (4.7).